Issues and Resolutions to Open Problems in *Rational Rotations of Three Dimensional Euclidean Space*

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Abstract

This companion records the mathematical findings of the constructive Lean formalisation of N. J. Wildberger's revised Euclidean rotation paper. It gives an explicit repair to the perpendicular-direction argument for null anchors, answers the associativity question in Problem 80, gives exact finite-field counts and conjugacy types for Problem 81, and states the finite-word and finite-order results obtained for Problem 82. It also supplies the determinant-weighted Clifford correspondence requested in Problem 83. Rational normalization, positive rays, projective proportions and finite formulations of size claims are kept distinct. Numbering refers to the revised source, not its earlier August draft. The remaining subgroup-classification and arithmetic questions are identified explicitly.

1 Source, conventions and status

The source is the supplied 33-page revision of *Rational rotations of three dimensional Euclidean space* [1]. Its numbered assertion inventory ends at Example 79, followed by Problems 80–83. We use row vectors,

$$B(v,u) = v \cdot u, \qquad Q(v) = v \cdot v, \qquad p(\rho) = \frac{3 - \operatorname{tr}\rho}{4}.$$

All general-field statements below assume characteristic different from two. An assertion involving an ordered field additionally assumes its ordinary ordered-field laws. The original quaternion product has the row-action signs

$$(x : v)(y : u) = (xy - v \cdot u : xu + yv - v \times u), \qquad N(x : v) = x^2 + Q(v). \tag{1}$$

The displayed representatives, their nonzero scales and their exact coefficients are retained throughout.

The revised paper already repairs several defects discussed for the August draft. In particular, Corollary 51 now requires both input caps to be defined, Theorem 54 states the cap domains needed for its proof of conditional associativity, and Problem 80 has fixed linear coefficients and excludes the zero cross coefficient. The old projection counterexample to the earlier four-coefficient conjecture is consequently not a counterexample to the new question. It should not be carried forward as one.

Source item	Result of this companion
Theorem 33	The formula is retained; a missing zero/null-anchor existence case in its printed proof is supplied.
Problem 80	Complete classification in characteristic different from two: $\delta = -\gamma^2$, using explicit scalar probes.
Problem 81	Exact counts, spinor classes and conjugacy types; one nonidentity parabolic conjugacy class in the full rotation group.
Problem 82	Exact finite-word spinor image, kernel/coset tests and finite-order table. Internal classification of generated subgroups remains open.
Problem 83	The projective Clifford correspondence for every displayed nonsingular symmetric ternary form.

This is an account of proved statements and their limits, rather than a claim to have solved every research question in the source. Scope boundaries concerning completed infinite objects and topology are not mathematical errors in the source.

2 Theorem 33: non-null perpendicular directions

Definition 32 permits a cap with a zero or null anchor. The last sentence of the proof of Theorem 33 appeals to nondegeneracy of the plane perpendicular to a *non-null* anchor. This establishes existence in that case, but does not cover the identity anchor or the nonzero null anchor in Example 45. The spread formula itself remains correct.

Lemma 2.1 (Four-candidate perpendicular construction)**.** *For every $v = (t, r, s) \in F^3$, one can explicitly find w with $w \cdot v = 0$ and $Q(w) \neq 0$.*

Proof. Test the three perpendicular vectors

$$w_1 = (-r, t, 0), \qquad w_2 = (-s, 0, t), \qquad w_3 = (0, -s, r).$$

Any non-null candidate suffices. If all three are null, then

$$Q(w_1) + Q(w_2) - Q(w_3) = 2t^2 = 0.$$

Since $2 \neq 0$, this gives $t = 0$. The vector $w = (1, 0, 0)$ is then perpendicular to v and has quadrance one. The construction includes $v = 0$, and each branch uses just a scalar equality test. □

For a cap $\rho = c(A_v)$ and such a w, quadratic reduction gives

$$w\rho = \frac{(1 - Q(v))w + 2(v \times w)}{1 + Q(v)}.$$

The cross term is perpendicular to w, and the rotation preserves its quadrance. Therefore

$$s(w, w\rho) = 1 - \frac{(1 - Q(v))^2}{(1 + Q(v))^2} = 4p(\rho)(1 - p(\rho)).$$

For a half turn the perpendicular vector is negated and the spread is zero. This proves independence of the chosen admissible direction in all charts. The proposed source repair is simply to replace the final existence sentence of Theorem 33 by the four-candidate construction.

3 Problem 80: the associative coefficient products

The revised question concerns

$$v \star u = \frac{v + u + \gamma(v \times u)}{1 + \delta(v \cdot u)}, \qquad \gamma \neq 0, \tag{2}$$

on its displayed nonzero-denominator domain. "Associative wherever defined" means equality of the two bracketings whenever all four products occurring in them are defined. It does not mean that the two bracketings have identical domains.

Theorem 3.1 (Answer to Problem 80)**.** *For $\gamma, \delta \in F$ with $\gamma \neq 0$, the operation* (2) *is associative wherever defined if and only if $\delta = -\gamma^2$.*

Proof. Let $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$. For any nonzero a with $D = 1 + \delta a^2 \neq 0$, all four products for the triple (ae_1, ae_1, e_2) are then defined. The e_2-coordinate of $(ae_1 \star ae_1) \star e_2$ is one, whereas the same coordinate of $ae_1 \star (ae_1 \star e_2)$ is

$$\frac{1 - \gamma^2 a^2}{1 + \delta a^2}.$$

Equality gives $(\delta + \gamma^2)a^2 = 0$, hence the claimed relation. The necessary probe or coefficient relation is obtained finitely. Try $a = 1$. If its denominator vanishes, then $\delta = -1$. If $\gamma^2 = 1$, the desired relation already follows. Otherwise take $a = \gamma \neq 0$, whose denominator $1 - \gamma^2$ is nonzero. This argument also covers characteristic three; no appeal to an infinite field is needed.

For sufficiency assume $\delta = -\gamma^2$. The invertible dilation $D_\gamma(v) = \gamma v$ transports $\star$, including its denominator, to the opposite anchor product:

$$D_\gamma(v \star u) = \frac{D_\gamma(v) + D_\gamma(u) + D_\gamma(v) \times D_\gamma(u)}{1 - D_\gamma(v) \cdot D_\gamma(u)}.$$

Its homogeneous product is the opposite of the associative quaternion product (1). Comparing affine representatives proves associativity on every common domain, even without assuming that all intermediate vectors are cap anchors. □

The fixed linear coefficients already make zero a two-sided identity. The nonzero cross coefficient in the revised question permits the two-probe argument above over every stated field. The retained formal theorem states the stronger homogeneous rational classification: for

$$\frac{\alpha v + \beta u + \gamma(v \times u)}{\epsilon + \delta(v \cdot u)},$$

zero is a two-sided identity and the product is associative wherever defined exactly when

$$\epsilon \neq 0, \qquad \alpha = \beta = \epsilon, \qquad \epsilon\delta = -\gamma^2.$$

Thus the revised normalized problem has a positive classification answer; the degenerate projection operations excluded by its new statement play no role in that answer.

4 Problem 81: finite fields

Let $F = \mathbb{F}_q$ with q odd, and write $\eta(c) = 1, -1, 0$ according as c is a nonzero square, a nonsquare, or zero. The constructions use a supplied complete finite carrier and its exact square table.

4.1 Vector counts and half-spread fibers

Lemma 4.1 (Ternary quadric count)**.** *For $a \in F$,*

$$\#\{v : Q(v) = a\} = q^2 + q\eta(-a).$$

Proof. Put $\varepsilon = \eta(-1)$, and let $C(b) = \#\{(r, s) : r^2 + s^2 = b\}$. If $-1 = i^2$, the substitution $(r, s) \mapsto (r + is, r - is)$ counts solutions of $xy = b$, giving $C(0) = 2q - 1$ and $C(b) = q - 1$ for $b \neq 0$. If -1 is nonsquare then $C(0) = 1$. For $b \neq 0$, the two square sets of size $(q + 1)/2$ give a point on the conic $r^2 + s^2 = bt^2$. Lines through it parametrize $q + 1$ projective points; none is at infinity, so $C(b) = q + 1$. In both cases

$$C(0) = q + (q - 1)\varepsilon, \qquad C(b) = q - \varepsilon \quad (b \neq 0).$$

Sum $C(a - t^2)$ over t. There are one, two or zero exceptional summands according as a is zero, a nonzero square or a nonsquare. These three counts give the stated formula. □

Theorem 4.2 (Rotation count and half-spread distribution)**.** *The complete rotation count is $q(q^2 - 1)$. Its half-spread fibers are*

$$\#\{\rho : p(\rho) = h\} = \begin{cases} q^2, & h = 0 \text{ or } h = 1, \\ q^2 + q\eta(-h(1 - h)), & h \neq 0, 1. \end{cases}$$

The two spinor-norm fibers each have $q(q^2 - 1)/2$ elements.

Proof. In the affine chart $(1 : v)$, exclude $Q(v) = -1$, which removes $q^2 + q$ vectors. At infinity the non-null lines number

$$\frac{(q^3 - 1) - (q^2 - 1)}{q - 1} = q^2.$$

Their sum is $q^3 - q$. For $h \neq 1$, the unique affine representative satisfies $Q(v) = h/(1 - h)$. Apply the lemma, observing that the square class of $-h/(1 - h)$ equals that of $-h(1 - h)$. Half turns are precisely the q^2 points at infinity. Every nonzero field element is a sum of two squares, so both norm classes have quaternion representatives. Multiplication by a representative of nonsquare norm bijects their rotation fibers. They consequently have equal size. □

For $h \neq 1$, every rotation in a half-spread fiber has spinor norm $[1 - h]$. For $h = 1$, its norm is the square class $[Q(v)]$ of its axis. More explicitly, the half-turn axes of either prescribed square class $[c]$ number

$$\frac{q^2 + q\eta(-c)}{2}.$$

Indeed, there are $(q - 1)/2$ nonzero quadrances in that class, each having the lemma's vector count, and each projective line contains $q - 1$ nonzero vectors.

4.2 Conjugacy types and parabolics

The reciprocal factor of the characteristic polynomial is

$$X^2 - 2(1 - 2h)X + 1, \qquad \Delta = -16h(1 - h).$$

For $h \neq 0, 1$, it has distinct roots. Square $-h(1-h)$ gives split semisimple type; nonsquare $-h(1-h)$ gives nonsplit semisimple type. Through the retained split-quaternion matrix model, there is one

projective conjugacy class for each such fixed h. Indeed, the matrix trace is nonzero when $h \neq 1$; scale it to one. The identity $1 - h = (\operatorname{tr} M)^2/(4 \det M)$ then fixes its determinant at $1/(4(1-h))$, so both matrices have the same characteristic polynomial. A non-scalar two-by-two matrix has a cyclic vector: test the two coordinate vectors and their sum. If none were cyclic, the first two tests would make the matrix diagonal and the third would force equal diagonal entries. In the basis consisting of a cyclic vector and its image, the matrix is the companion matrix of its characteristic polynomial. The two matrices are therefore similar. A half turn instead has a non-scalar trace-zero representative. Representatives with the same determinant square class can be scaled to the same determinant and the same cyclic-basis argument applies. The two determinant classes thus give its two conjugacy classes.

The zero fiber has characteristic polynomial $(X-1)^3$. Its identity has anchor zero; the other $q^2 - 1$ rotations have nonzero null anchors. There are $q + 1$ null axis lines and $q - 1$ nonzero parameters over each. These nonidentity parabolics form one conjugacy class in the *full* group $\mathrm{SO}(3, F) \simeq \mathrm{PGL}_2(F)$. To see this, a nonidentity unipotent projective matrix can be put in the form $\left(\begin{smallmatrix} 1 & a \\ 0 & 1 \end{smallmatrix}\right)$, $a \neq 0$; diagonal conjugation changes any such a to any other. This statement is about the full group, not conjugacy restricted to its index-two unit-representable subgroup.

4.3 The distribution on one non-null axis

Fix v with $Q(v) \neq 0$. The source's condition $Q(v)h(1-h)$ square for $h \neq 0, 1$ holds for exactly

$$\frac{q - 2 - \eta(-Q(v))}{2} \tag{3}$$

values. This is the count of points with $y \neq 0$ on

$$(h - \tfrac{1}{2})^2 + Q(v)y^2 = \tfrac{1}{4},$$

after removing its two points with $y = 0$ and pairing the remaining points as $\pm y$. Thus the count is $(q-3)/2$ if $-Q(v)$ is a square, and $(q-1)/2$ otherwise. Each amount has exactly two inverse cap rotations. The forbidden affine scales are counted by $\lambda^2 Q(v) = -1$, equivalently the square-root fiber of $-Q(v)$. These are actual filtered-carrier and root-fiber counts in the formalisation, including the literal examples over $\mathbb{F}_3$ and $\mathbb{F}_5$.

As a check on the global distribution, the half-spread counts for $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$, in the order of the field's integer representatives, are respectively

$$(9, 9, 6), \quad (25, 25, 20, 30, 20), \quad (49, 49, 56, 42, 42, 42, 56).$$

They sum to $24,120,336$. Over $\mathbb{F}_5$ there are additionally 36 null quaternion proportions, 6 null axis lines and 24 nonidentity parabolics, agreeing with Example 70.

5 Problem 82: what the spinor norm determines

5.1 Normalization and the kernel

For a rational nonzero quaternion representative P, $N(P) > 0$, and

$$\chi(\rho(P)) = [N(P)].$$

A rational norm-one representative exists exactly when this class is trivial: $N(P) = r^2 \neq 0$ permits division by r, while any unit rescaling exhibits such a square. The rotations with this property form $K = \ker \chi$, closed under products and inverses. For a half turn the criterion is exactly that its supplied axis quadrance is a square.

5.2 Finite generator lists and signed words

Theorem 5.1 (Complete finite spinor image)**.** *Given a finite list of rational rotations $g_1, \dots, g_m$, one can compute their complete finite spinor image, a word realizing every image element, and exact kernel and kernel-coset tests for every supplied finite signed word.*

Proof. For positive $c = a/b$ in reduced form, let $S(c)$ be the finite set of primes occurring to odd exponent in ab. Finite factorization gives

$$[c] = [d] \iff S(c) = S(d), \qquad S(cd) = S(c) \triangle S(d).$$

Choose retained representatives for the generators and take the union S of their supports. Their parity vectors $s_i \in \mathbb{F}_2^S$ span a finite space V. The spinor vector of a signed word is the sum of its generator vectors; inverses have the same vector. Thus a word is in K exactly when its vector is zero, and two words lie in the same K-coset exactly when their vectors agree. Finite binary elimination computes the rank r and the 2^r image vectors, retaining generator words realizing them. The quaternion product separately evaluates each word's exact rotation. □

This classifies the image and the quotient detected by χ, not the generated subgroup internally. Different words can have the same spinor vector and different rotation matrices. Matrix comparison decides any supplied pair, but does not provide an isomorphism type, presentation or complete relation theory of the generated subgroup.

5.3 Finite-order values

Theorem 5.2 (Rational finite-order table)**.** *If $\rho \in \mathrm{SO}(3, \mathbb{Q})$ is supplied with an exact positive order, then the following table is exhaustive:*

order	$p(\rho)$	$\chi(\rho)$
1	0	[1]
2	1	$[Q(a)]$ *for its nonzero axis* a
3	3/4	[1]
4	1/2	[2]
6	1/4	[3].

Proof. Set $\tau = \operatorname{tr}\rho - 1$ and use the finite recurrence

$$L_0 = 2, \quad L_1 = \tau, \quad L_{k+2} = \tau L_{k+1} - L_k.$$

Cayley–Hamilton gives $\operatorname{tr}(\rho^k) = 1 + L_k$. An identity power implies $L_n = 2$, and the monic integer equation $L_n(\tau) - 2 = 0$ forces a reduced rational τ to be integral by denominator clearing and Bézout. The Euclidean half-spread bounds give $-2 \le \tau \le 2$. The five integer values produce exactly the orders in the table; the retained smaller nonidentity powers certify minimality. Finally $p = (2 - \tau)/4$, and $\chi = [1 - p]$ away from half turns, give its last two columns. At a half turn use $(0 : a)$ instead. □

The order-two row is a complete answer for a supplied axis. A separate global classification of the positive square classes represented by three rational squares has not been supplied. No claim that every positive square class occurs as an order-two value is made here.

5.4 Retained finite-precision approximation

The earlier development also proves a precise finite approximation statement: for each rational rotation ρ and positive rational ε, it constructs $\eta \in K$ with

$$\sum_{i,j}(\eta_{ij} - \rho_{ij})^2 < \varepsilon.$$

Choose the projectively equivalent sign of $P = (x : v)$ with $x \geq 0$. Put $n = N(P) > 0$, choose a sufficiently tight positive rational upper square-root bracket endpoint u, and set

$$A = u + x, \quad R = (A^2 - Q(v) : 2Av), \quad D = A^2 + Q(v), \quad U = D^{-1}R.$$

The retained construction ensures $D \neq 0$. Exact expansion gives $N(R) = D^2$, hence $N(U) = 1$, and the rotation of U satisfies

$$\sum_{i,j}(\eta_{ij} - \rho_{ij})^2 = \frac{8(u^2 - n)^2 Q(v)}{D^2 n}.$$

Finite integer square-root refinement makes this less than the requested bound. This is a theorem for each finite request. It neither constructs a completed topological space nor asserts a formal closure theorem.

6 Problem 83: arbitrary nondegenerate ternary forms

Let M be a supplied symmetric nonsingular three-by-three matrix over F, and put

$$B_M(u, v) = uMv^T, \quad Q_M(u) = B_M(u, u), \quad d = \det M, \quad C_M(u, v) = (u \times v)\operatorname{adj}(M).$$

The cross on the right is the ordinary coordinate cross product. It is essential to retain the determinant factors: a bare replacement of the Euclidean dot product would give the wrong algebra for a general form.

Theorem 6.1 (Form-relative projective Clifford correspondence)**.** *The product and norm*

$$(x : u)(y : v) = (xy - dB_M(u, v) : xv + yu - C_M(u, v)), \qquad N_M(x : u) = x^2 + dQ_M(u)$$

are associative and multiplicative, with

$$P\bar{P} = \bar{P}P = (N_M(P) : 0), \qquad \overline{(x : u)} = (x : -u).$$

For $N_M(P) \neq 0$, normalized conjugation

$$(0 : w) \longmapsto N_M(P)^{-1}\bar{P}(0 : w)P$$

is a member of $\mathrm{SO}(M, F)$*. Every member occurs, and two non-null representatives give the same rotation if and only if they differ by a nonzero scalar.*

Proof. Adjugate identities give the determinant-weighted Lagrange, Binet–Cauchy and vector triple-product identities. Expanding the displayed finite coordinates proves associativity, the conjugation identities and norm multiplicativity. The same identities prove preservation of B_M and determinant one of the action.

For surjectivity, the retained constructive two-reflection theorem gives two non-null normals for any supplied rotation. Their pure Clifford representatives give half turns; multiplying the two half turns gives the same rotation as multiplying the reflections. This constructs a non-null preimage, including for the identity. The construction selects non-null vectors and displacements from explicit finite coordinate candidates. For faithfulness, the action identities force a trivially acting non-null representative to have vector part zero. Applying this to the relative product of two representatives and cancelling the non-null factor gives the nonzero scalar witness. Conversely a rescaling multiplies both action numerator and norm by the same square. □

The affine chart $(1 : u)$ has

$$p = \frac{dQ_M(u)}{1 + dQ_M(u)}, \qquad u \star_M v = \frac{u + v - C_M(u, v)}{1 - dB_M(u, v)}.$$

For defined input caps and nonzero product denominator,

$$1 + dQ_M(u \star_M v) = \frac{(1 + dQ_M(u))(1 + dQ_M(v))}{(1 - dB_M(u, v))^2}.$$

Thus all the affine exceptional loci and their projective resolution extend with the same retained form. No diagonalizing basis or square-root normalization is required. The form I gives the Euclidean algebra; $\mathrm{diag}(1, -1, -1)$, with the source's explicit vector sign conversion, gives the dihedron convention of the Minkowski companion [2].

7 Rays, normalization and the size boundary

A quaternion ray identifies P with aP only for $a > 0$. A projective proportion permits every $a \neq 0$, so it consists of two opposite rays. Quaternion multiplication and inversion respect positive scaling; over an ordered field $P^{-1} = \bar{P}/N(P)$ is in the same ray as $\bar{P}$, because $N(P) > 0$. Equality of rotation matrices means the representatives are projectively equivalent, hence either on the same ray or on opposite rays. Each rotation fiber therefore contains exactly two distinct rays; the kernel consists of the scalar rays of 1 and -1.

These quaternion rays must not be confused with rays of the *anchor vector*. The quaternions P and $-P$ give the same rotation. The nonzero affine anchors v and $-v$ give inverse rotations, and are distinct vector rays. The source's orientation data concerns this second distinction.

A rational quaternion ray has a unit representative precisely when its norm class is trivial. When it exists, its unit representative is unique within that positive ray. Thus the full rational ray cover is strictly larger than the rational norm-one cover. The Pauli matrix description and its algebraic determinant-one unitary image are checked using a formal imaginary scalar; no completed complex field or Lie group is needed for those identities. The real spin-group and trigonometric identifications lie outside the present formal constructive scope.

The infinitude clause of Theorem 74 and the infinite-index clause of Corollary 75 have the following checked finite content. For every supplied natural number n, an executable construction returns n pairwise inequivalent positive rational square classes and rotation preimages, and n rotations pairwise separated modulo K. Successive finite prime constructions supply the classes; a four-square construction supplies exact quaternion norms. The pairwise certificates are retained. This gives arbitrary finite size without constructing an infinite quotient carrier.

An additional retained result, no longer a numbered open problem in the revision, is point–plane polarity: $[x]$ corresponds to the plane with normal x. The identity $(x\rho) \cdot (y\rho) = x \cdot y$ makes this

correspondence equivariant and preserves incidence. For non-null representatives, projective point quadrance equals spread of the corresponding plane normals. Null representatives retain incidence but do not admit that divided spread.

8 Formal evidence and remaining questions

The reusable mathematical owners remain in `Purelib`; the publication facade is `formalisations/RationalRotations/RationalRotations.lean`. Its claim ledger is `formalisations/RationalRotations/coverage.md`, with the earlier draft's numbering preserved separately in `formalisations/RationalRotations/coverage-2026-08-31.md`. The following map gives direct entry points rather than copied proofs.

Result	Mathematical owner / retained entry point
Perpendicular witness	`Purelib/Geometry/Quadratic/Space3/Perpendicular.lean`; `perpendicularNonNullWitness`.
Coefficient classification	`Purelib/Geometry/Rotations/Cayley/CoefficientProducts.lean`; `rational_hardenedAssociativityClassification`.
Finite strata and conjugacy	`Purelib/Geometry/Rotations/FiniteField/Classification.lean`; the original finite-field presentation and its classification certificate.
Fixed-axis amounts	`Purelib/Geometry/Rotations/Axis/Finite.lean`; `admissibleAmounts_length`, cached roots and two-scale fibers.
Generated spinor image	`Purelib/Geometry/Rotations/Rational/Generators/Classification.lean`; `GeneratedSubgroups.classify`.
Finite orders and approximation	`Purelib/Geometry/Rotations/Rational/Order.lean`; `Purelib/Geometry/Rotations/Rational/Kernel/Approximation.lean`.
General ternary correspondence	`Purelib/Geometry/Rotations/Ternary/Theory.lean`; `TernaryForm.generalRotationTheory`, together with `projectiveCliffordCorrespondence`.
Ray cover	`Purelib/Geometry/Rotations/Projective/RayCover.lean`; the original `RayProportion` product, inverse and fiber laws.

The mathematical tree at commit `a6219f044` passed all twelve completion gates on 11 October 2026: warning-fatal full build and lint, the full configured-script axiom audit, independent nanoda checking and the tooling regressions. Builds used 64 Lean threads. The normal proof-closure audit accepted 136,660 project declarations and 1,536 selected canonical declarations with no allowed axioms; nanoda checked all 42 compatible module sets. This certifies the selected mathematical proof closure, not a separate strict audit of every internal meta-provider helper. These papers change documentation only; they do not introduce new Lean declarations. The finite-field counts, stratum certificates and parabolic conjugators have retained Lean proofs. The cyclic-matrix argument above additionally deduces the full semisimple and half-turn conjugacy classes in this paper; it is not claimed as a newly checked standalone Lean classification theorem.

Problem 80 is answered over every field of characteristic different from two, and the finite-field and general-form results above answer Problems 81 and 83 in the displayed constructive sense. Problem 82 remains open as an internal classification of natural generated subgroups. A global list of order-two spinor classes and any completed topological statement would need additional results. Historical priority and numerical-performance claims have not been inferred from the formalisation.

References

[1] N. J. Wildberger, *Rational rotations of three dimensional Euclidean space*, supplied revision `RationalRotationsQuaternions2.pdf`, 33 pages. Source SHA-256: `3e03d449b480ab4b27b520e8059495bab2e62cdf3852b1ff33e02a38c7ad8c53`.

[2] *Issues and Resolutions to Open Problems in Rational Rotations of Three Dimensional Minkowski Space*, companion in `formalisations/MinkowskiRotations/IssuesAndOpenProblems.pdf`, 12 October 2026.